

This model uses **Elo-style team ratings** (not official FIFA rankings) to simulate the World Cup and predict:

- group winners
- teams advancing
- tournament champions
- Golden Boot winners

Match outcomes are simulated using Elo-based win probabilities and Monte Carlo tournament simulations.

Team Ratings

Higher rating = stronger team.

Example scale:

- 2000+ = elite World Cup contender
- 1900s = strong contender
- 1800s = competitive national team
- 1700s = solid World Cup-level team

The ratings can later be improved using:

- FIFA rankings
- Elo ratings
- betting markets
- historical match data
- FBref player stats

Roster Adjustments

Teams get small boosts based on squad quality, such as:

- elite club players
- top European league players
- squad depth

Tournament Simulation

The model simulates the tournament many times to estimate:

- chance to win the group

- chance to advance
- chance to win the World Cup

Golden Boot Model

Player goal predictions depend on:

- scoring rate
- expected minutes
- starter probability
- penalty duties
- team advancement
- group difficulty

Example:

Players on stronger teams usually get more matches, which increases scoring chances.

Current Strengths

- strong tournament simulation structure
- realistic advancement logic
- accounts for group difficulty
- considers player availability/injuries

Current Weaknesses

- team ratings are derived from World Football Elo ratings and adjusted using roster and player-production metrics
- player scoring assumptions are hand-entered
- FBref data has not been fully integrated yet

Future Improvements

- integrate live FBref player production data
- improve injury and roster availability tracking
- refine knockout-stage simulation logic
- incorporate betting market implied probabilities

Data Sources

- World Football Elo Ratings
- FBref player statistics
- historical World Cup data
- roster availability adjustments